In Llama, Saying "I Don't Know" Tracks Entity Popularity More Than Knowledge

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Abstract

We investigate what a language model's "I don't know" tracks: whether the model lacks the fact, or how popular the entity is. We cross entity popularity (top vs. bottom decile of Wikipedia page views in PopQA) with knowledge labels taken from the model's own forced answers. We find that in Llama-3.1-8B-Instruct, saying "I don't know" tracks popularity more than knowledge. On obscure entities, Llama says "I don't know" in 43% of replies to questions whose greedy answer and at least 9 of 10 forced samples are correct; on popular entities, in 0%. The gap replicates with fresh replies and in new questions, and survives a control that removes facts recoverable from the name alone. Its size depends on the abstention measure (+0.13 to +0.44) and falls to +0.14 [−0.01, +0.28] when all label corrections are applied at once, while the popularity gap among facts Llama does not know (+0.34) survives every check. Gemma-2-9B shows the same preference when the prompt permits "I don't know"; for three other families the judges disagree too much to measure it. Steering and logit-lens readouts point to a generic abstention signal, not a familiarity mechanism.

1 Introduction

Saying "I don't know" when a fact is missing avoids the worst hallucinations [Kalai et al., 2025]. Confidence estimates usually read uncertainty over answers [Kadavath et al., 2022, Kuhn et al., 2023, Stolfo et al., 2024], and models use such confidence when a prompt offers abstention [Kumaran et al., 2026]. However, verbal and answer-level uncertainty diverge [Ji et al., 2025], so what a model's "I don't know" actually tracks is less clear.

One candidate is familiarity. In people, the sense of being able to answer depends on how familiar a question's terms are as well as on whether the answer can be retrieved [Reder and Ritter, 1992, Schwartz and Metcalfe, 1992, Koriat and Levy-Sadot, 2001]. In language models, "known entity" features suppress a default "can't answer" response. Lindsey et al. [2025] propose that a familiar name can trigger these features when the fact is missing, so the model answers when it should abstain. We call this a *misfire*. The converse failure is an unfamiliar name that does not suppress the default: the model then abstains on a fact it knows.

Testing this requires separating familiarity from knowledge. Existing tests define an entity as known when the model answers about it correctly [Ferrando et al., 2025], so the two move together. Familiarity has been separated from factual reliability in representations [Brzezinka, 2026a] and shown to shape hallucination through fine-tuning data [Kang et al.,

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2025], but, to our knowledge, not crossed with knowledge in abstention behavior. We use page-view popularity as the measurable stand-in for familiarity.

We take questions about obscure and popular entities from PopQA [Mallen et al., 2023] and label each fact known or unknown from the model's own answers [Gekhman et al., 2024]. If abstention follows popularity, two cells show it: popular unknown facts should draw few "I don't know" replies (misfires), and obscure known facts many (the default not suppressed). In Llama, they do (Figure 1, Table 1). This also bears on representational convergence [Huh et al., 2024]: do models that agree on answer uncertainty also agree on what makes them abstain? Of five families, only Llama and Gemma let us measure it.

Main Contributions.

1. **Popularity Over Knowledge in Llama:** a popularity preference of +0.30 [+0.20, +0.40] that replicates with fresh replies (+0.29) and in new questions (+0.10), and survives a name-cue control (+0.36; Section 3.1). Among unknown facts alone, the popularity gap is +0.34 [+0.25, +0.43].
2. **Same Preference in Gemma When Permitted:** +0.30 [+0.19, +0.40] under a permissive prompt (Section 3.2); the judges cannot measure Qwen, Mistral or OLMo-2.
3. **Generic Abstention, No Familiarity Mechanism:** steering along obscure-minus-popular adds nothing beyond an abstention direction fitted on popular questions only (Section 3.3).

2 Setup

Questions & Labels. We use the 16 relations of PopQA. Within each relation, subjects in the bottom and top deciles of Wikipedia page views form the *obscure* and *popular* arms. For each model, we grade a greedy answer and ten samples ($T = 0.7$) against gold aliases, under a *forced-answer prompt* (the chat template plus the assistant prefix "The answer is"). A fact is *known* if the greedy answer and at least 9 samples are correct, and *unknown* if neither the greedy answer nor any sample is; the band between is dropped. "Known" is thus specific to this prompt. This leaves 2,303, 2,392, 2,282 and 2,378 questions for Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Gemma-2-9B-it and Mistral-7B-Instruct-v0.3 [Grattafiori et al., 2024, Qwen et al., 2024, Team et al., 2024, Jiang et al., 2023] (cell counts in Appendix A).

Two Readouts. *Answer uncertainty* is the first-token entropy under the forced-answer prompt. The knowledge labels come from the same distribution, so entropy should follow knowledge; it serves as a control. *Abstention* is measured with a *free prompt* (the chat template alone): for up to 100 questions per popularity × knowledge cell (387 in total), we draw three replies ($T = 0.7$, up to 96 tokens). Two LLM judges (Qwen2.5-32B-Instruct, Gemma-2-27B-it) label each reply DONT_KNOW, CLARIFY (asks which entity is meant) or ANSWER; a reply abstains when both say DONT_KNOW. On Llama the judges agree at Cohen's $\kappa = 0.62$ [0.57, 0.67], above a pre-specified bar of 0.60 for the point estimate; on the other families they do not. A *permissive prompt* ends the question with "If you do not know the answer, say that you do not know."

Effects. The popularity effect is the obscure-minus-popular difference in a readout, averaged over the two knowledge levels; the knowledge effect is the unknown-minus-known difference, averaged over popularity. Their difference is the model's *popularity preference*: positive when abstention follows popularity more than knowledge. In this equally weighted 2×2, it equals abstention on obscure known facts minus abstention on popular unknown facts. Intervals are 95% bootstraps over questions within cells unless stated; because arms are defined within relations, we also report relation-clustered intervals. Unless marked post hoc, every analysis followed a plan written and hashed before it ran (*pre-specified*; Appendix G).

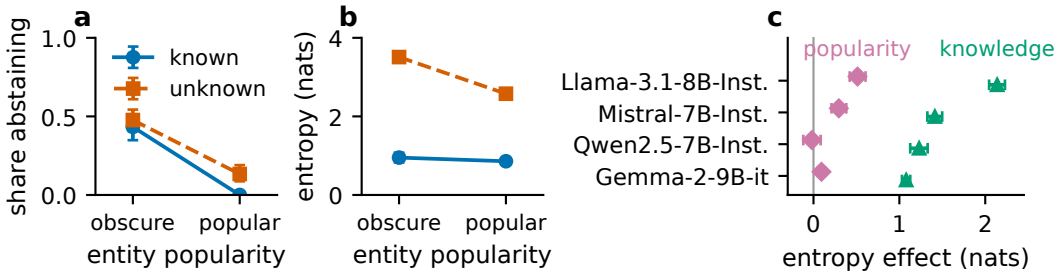


Figure 1: **In Llama, saying "I don't know" tracks popularity; answer entropy, a control, follows knowledge.** (a) Share of Llama's free replies both judges label "I don't know"; (b) its first-token entropy when the reply must begin "The answer is" (values and n in Table 1); (c) popularity and knowledge effects on that entropy in four instruct models. 95% bootstrap intervals.

Table 1: Llama-3.1-8B-Instruct by cell: share of free replies both judges label "I don't know", and forced-answer first-token entropy, with 95% intervals; counts are for abstention, and entropy uses the full pool (Appendix A).

	Obscure		Popular	
	known	unknown	known	unknown
Says "I don't know"	0.43 [0.35, 0.51]	0.48 [0.41, 0.55]	0.00 [0.00, 0.00]	0.13 [0.08, 0.19]
questions	87	100	100	100
Entropy (nats)	0.95 [0.83, 1.08]	3.51 [3.46, 3.56]	0.86 [0.80, 0.92]	2.58 [2.47, 2.69]

3 Results

3.1 Llama: Abstention Tracks Popularity More Than Knowledge

Core Result. On obscure entities whose facts it knows, Llama says "I don't know" in 0.43 of replies; on popular entities whose facts it knows, it never does (Table 1). Its forced-answer entropy is similar in the two cells (0.95 vs. 0.86 nats). The popularity effect on abstention is +0.39 [+0.33, +0.45] and the knowledge effect +0.09 [+0.03, +0.15]: a popularity preference of +0.30 [+0.20, +0.40]. Its sign holds under six other measures of abstention, from +0.13 to +0.44 (Table 2); we adopted the primary measure after the first pre-specified one returned the smallest of these.

Replications & Name-Cue Control. With fresh replies, the preference is +0.29 [+0.20, +0.38]. In new questions from the 10th–30th and 70th–90th page-view percentiles, it is smaller but positive (+0.10 [+0.01, +0.20]), as the narrower contrast predicts. Are obscure "known" facts just guessed from the name? We re-asked each known question with three pseudo-names (two letters swapped, one vowel changed). Llama still answered 34 of 86 obscure known facts correctly (25 of 46 country questions); these abstain less (0.33) than the rest (0.50), and without them the preference is +0.36 [+0.25, +0.48].

Label & Measurement Checks. Relabeling knowledge with a paraphrased question agrees with the original labels at $\kappa = 0.79$ (on the 253 questions labeled under both); 27 of 87 obscure known facts stay known and abstain at 0.46, and the preference is +0.32 [+0.16, +0.49]. A larger model (Llama-3.3-70B) regraded popular unknown facts by meaning and moved 9 of 100 to known and 31 to the dropped band; the preference becomes +0.23 [+0.12, +0.35]. With a third judge it is +0.44; counting only replies that decline without naming any candidate, +0.43 [+0.31, +0.54]. At most 0.03 of "I don't know" replies in any cell contain the gold answer (post hoc). All label corrections at once (post hoc) leave +0.14 [−0.01, +0.28], while the gap among unknown facts stays +0.18 [+0.04, +0.31].

Relation Concentration. Llama knows few obscure facts: 80 of the 87 come from three relations (capital, country, sport), as do 85 of 100 in the new questions. With relations as the

resampling unit, the preference's interval is [−0.10, +0.43] (post hoc; leave-one-relation-out +0.24 to +0.35), while the gap among unknown facts stays clear ([+0.23, +0.46]). The comparison of known facts therefore rests on a few relations.

Recognition, Ambiguity & Familiarity. Entity recognition (LLM-graded) and Wikidata name collisions do not explain the gap (Appendix A). But no model-side measure shows the obscure names to be unfamiliar to Llama (Appendix F), so the result has the shape the misfire idea predicts without establishing its mechanism.

Within One Prompt: Mixed. The pre-specified readout at the forced-answer position, the first token's lean toward hedge tokens (refusal openers such as " I", " not", " unknown"), shows no preference (−0.12 [−0.72, +0.45]; Appendix B). A reading at the free prompt, specified after seeing this, shows one: popularity raises the log-odds of a first token "I" more than knowledge does (difference +5.3 [+4.4, +6.0]), while knowledge raises its entropy more (−0.11 [−0.18, −0.03] nats).

3.2 Other Families: Gemma Shares the Preference When Permitted

Plain Prompt: Too Few Abstentions to Measure. Entropy follows knowledge in all four families (Figure 1c), but Qwen, Gemma and Mistral rarely say "I don't know"; Gemma abstains in at most 0.11 of replies per cell, so its preference cannot be estimated. A pre-specified four-family test found partial support for Qwen and Mistral, and full support for Gemma on a readout that shows no preference in Llama (Appendix B).

Permissive Prompt: Same Preference in Gemma. With the permissive prompt, Llama and Gemma abstain often and three judges agree (Fleiss $\kappa = 0.71$ and 0.67). The popularity preference is +0.47 [+0.35, +0.59] for Llama and +0.30 [+0.19, +0.40] for Gemma, which says "I don't know" to 0.41 of obscure known and 0.00 of popular known questions. For Qwen, Mistral and OLMo-2-7B, the judges disagree (Fleiss κ 0.40–0.50), so their preferences are unmeasured (point estimates +0.15, +0.16, −0.09; Appendix B). Cross-model agreement on the forced-answer hedge readout met a post hoc rule for one of three pairs (Appendix E).

3.3 Mechanistic Tests: No Familiarity Cause Isolated

Steering: A Generic Abstention Signal. Like work steering refusal along recognition directions [Ferrando et al., 2025, Brzezinka, 2026b], we fitted obscure-minus-popular, unknown-minus-known and abstain-minus-answer directions on half of the questions [Arditi et al., 2024, Rimsky et al., 2024], chose the dose on a second set and scored a third (Appendix C). In Llama, obscure-minus-popular adds +0.09 [+0.06, +0.12] abstention over random directions (short of the pre-specified +0.10 bar). It is nearly parallel to abstention directions fitted without any popularity contrast: cosine 0.91 for one fitted on obscure questions only and 0.89 for one fitted on popular questions only. Removing either leaves +0.01 [+0.00, +0.03] and +0.01 [−0.00, +0.02]. It beats unknown-minus-known on abstention (+0.06), as the cells' rates predict; in Gemma, unknown-minus-known raises entropy +0.66 [+0.58, +0.74] nats more (Figure 2b; cf. Che and Li, 2026).

Lens: Both Factors in Base Models. With prefitted Jacobian lenses [Gurnee et al., 2026], read at the forced-answer position in a pre-specified mid-depth window, popularity raises the hedge-token term in Llama-3.1-8B, Gemma-2-9B, Gemma-2-2B and OLMo-3-7B [Olmo et al., 2025], and knowledge raises it comparably or more in three of four (Appendix D, Figure 3). Instruct partners lean further toward knowledge, so this readout does not show where Llama-Instruct's preference arises.

4 Limitations

Abstention is judged by LLMs not validated against human labels, whose errors could differ by cell. "Known" is specific to the forced-answer prompt and concentrated in three relations; the design contrasts extreme deciles; models have 2–9B parameters. Hedging driven by ambiguity that the judges do not label as clarification is not excluded.

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A Robustness of the Llama abstention result

Design counts. Llama's pool has 87 known, 85 intermediate and 1179 unknown obscure facts, and 676, 285 and 361 popular ones; the two-judge sample is a stratified random sample of 100 questions per cell (seed 0), or all of a smaller cell. Intervals for differences of cell means use 10,000 within-cell bootstraps (seed 0); linear probability models use 2,000 bootstraps over questions.

Measurements. The single corrected judge gives the smallest difference. Its 24-token replies are often cut off before the model commits; why it and the regex disagree on the same replies is not established. The two-judge measurement with 96 tokens was designed for that reason after the +0.13 was seen and made primary before it ran. Regex recall on the 96-token replies, against the two-judge labels, is 0.99. Each judge's agreement with the three-judge majority: Qwen 0.98, Gemma 0.84, Llama-3.3-70B 1.00.

Table 2: Llama's popularity preference under other measures of abstention and other knowledge labels, and the coefficient of the obscure arm under regression controls. Positive values mean abstention follows popularity more than knowledge. "Corrected": regex rates on 24-token replies corrected for the recall and precision of one LLM judge (Qwen2.5-32B-Instruct), measured on a subset labeled by that judge, not by people. Plan: pre-specified (pre) or post hoc. n: questions.

Variant	Replies	n	Plan	Estimate
Measures of abstention: popularity preference				
Two judges, both DONT_KNOW (primary)	3, 96 tokens	387	pre	+0.30 [+0.20, +0.40]
Two judges, DONT_KNOW or CLARIFY	same	387	pre	+0.31 [+0.21, +0.41]
Three judges, majority	same	387	pre	+0.44 [+0.32, +0.55]
70B judge, declines naming no candidate	same	387	pre	+0.43 [+0.31, +0.54]
Regex detector	same	387	post hoc	+0.43 [+0.31, +0.55]
Regex detector	20, 24 tokens	2,303	pre	+0.44 [+0.34, +0.53]
One LLM judge, corrected	same 24-token	2,303	pre	+0.13 [+0.03, +0.24]
Replications and name-cue control (two judges): popularity preference				
Fresh replies, same questions	3, 96 tokens	387	pre	+0.29 [+0.20, +0.38]
New questions, 10th–30th vs. 70th–90th percentile	same	400	pre	+0.10 [+0.01, +0.20]
Name-recoverable known facts removed	same		pre	+0.36 [+0.25, +0.48]
Other knowledge labels (two judges): popularity preference				
Relabeled with a paraphrased prompt	3, 96 tokens	253	pre	+0.32 [+0.16, +0.49]
where both labelings agree	same		post hoc	+0.31 [+0.15, +0.48]
Unknown facts regraded by meaning	same		pre	+0.23 [+0.12, +0.35]
Popular unknown with the answer in a reply dropped	same		post hoc	+0.19 [+0.07, +0.31]
Guessable answers removed	same		post hoc	+0.30 [+0.19, +0.41]
All label corrections jointly[b]	same		post hoc	+0.14 [−0.01, +0.28]
Regression controls (two judges): coefficient of the obscure arm				
Relation fixed effects and knowledge only	3, 96 tokens	387	pre	+0.35
plus entity recognition	same	384	pre	+0.34 [+0.27, +0.41]
plus namesake count	same		post hoc	+0.34 [+0.27, +0.41]

[b]Meaning-based regrade of unknown facts, popular unknown questions with the gold answer in any free reply dropped, and guessable facts removed. Relation-clustered 95% intervals (resampling the 16 relations; post hoc): primary [−0.10, +0.43], name-cue control [−0.03, +0.48], new questions [−0.12, +0.27]; popularity gap among unknown facts +0.34 [+0.23, +0.46].

Entropy control. Forced-answer first-token entropy rises +2.14 nats when Llama lacks the fact and +0.51 when the entity is obscure (Figure 1c gives the four families).

Independent labels (pre-specified). Knowledge was relabeled with one paraphrased question per relation, the assistant prefix "Answer:" and fresh samples, under the same rule. Of the 253 questions labeled under both prompts, 62 are known under both, 169 unknown under both, 15 known only originally and 7 only under the paraphrase (agreement 0.91, $\kappa = 0.79$); most other changes move popular known facts to the dropped band. Where both labelings agree, cells have 99 / 27 / 70 / 35 questions (obscure unknown, obscure known, popular unknown, popular known). Under the new labels, the popularity effect on abstention is +0.38 [+0.29, +0.47] and the knowledge effect +0.06 [−0.03, +0.15], and original-prompt entropy follows them (knowledge minus popularity +1.40 [+1.02, +1.73]).

Regraded unknown facts (pre-specified). Llama-3.3-70B graded each of a fresh greedy and ten sampled forced answers for every unknown fact as naming the gold entity or not. Of the 100 popular unknown facts in the sample, 60 stay unknown, 9 become known and 31 intermediate (dropped); of the 100 obscure unknown facts, 86, 2 and 12. Dropping instead

every popular unknown question with a gold alias in any free reply (0.48 of them) gives +0.19 [+0.07, +0.31].

Strict declines (pre-specified). The 70B judge labeled each reply as declining without naming a candidate, hedging with a candidate, answering, or asking which entity is meant. Declines occur in 0.66 of obscure known replies and 0.00 of popular known ones; among replies both judges call DONT_KNOW, at most 0.03 per cell are hedged answers.

Entity recognition (pre-specified). Llama completed "{subject} is" after the question "What is {subject}?" (greedy and five samples), and Llama-3.3-70B graded each completion against the subject's English Wikidata description; recognition is the share of completions graded correct. It averages 0.83 for obscure known facts and 0.75 for popular known facts. Its correlation with log page views over all graded questions is $\rho = +0.52$. In the linear probability model its coefficient is -0.03 [-0.06, $+0.00$] per standard deviation, and it shrinks the popularity coefficient by 5% on the two-judge measure and 3% on the regex measure.

Guessable answers (post hoc). An answer counts as guessable if a gold alias of at least three characters appears in the subject's name, or if it is its relation's most common answer and covers at least 20% of the relation's pool. By this rule 30 of the 87 obscure known facts are guessable; they abstain at 0.42, and the other 57 at 0.44. Among unknown facts alone the popularity gap in abstention is +0.34 [+0.26, +0.43]. On the regex measure over the 2,106 non-guessable questions the difference is +0.48 [+0.36, +0.59].

Namesakes (post hoc). For each subject we counted Wikidata entities with the same English label; medians are 1 (obscure known), 5 (obscure unknown), 11 (popular known) and 13 (popular unknown). The covariate is the standardized log count. The 73 obscure known facts with a unique name abstain at 0.43.

Free replies (post hoc). Counting a gold alias only when it is not part of the subject's name, 0.25 of the free replies in the obscure known cell contain the answer, as do 0.44 of the replies that do not abstain and 0.00 of those that do (popular known: 0.98). For 9 of 87 obscure known questions, one free reply gives the answer and another says "I don't know". (An earlier version of this check also counted aliases inside the subject's name and overstated these shares.) A linear probability model of abstention gives popularity +0.29 [+0.23, +0.36], knowledge -0.05 [-0.14, $+0.03$] and +0.14 per standard deviation of forced-answer entropy.

Name spelling (post hoc). Between 55 matched pairs of known facts, one obscure and one popular, the regex abstention gap of +0.76 shrinks to +0.43 [+0.18, +0.69] once the names' word frequency, token count and token rarity are held equal. Over the 2,303 labeled questions, the popularity coefficient on the regex measure moves from -1.87 to -1.79 logits per standard deviation after adding the name's word frequency, token count and token rarity. The matched pairs compare only known facts, one popular with one obscure fact of similar surface form, which is why they lose more of the gap.

B Abstention and the forced-answer hedge readout in all four families

Intervals for the popularity-minus-knowledge rows: three judges, Llama [+0.32, +0.55], Gemma [-0.03, $+0.06$]; hedge log-odds, Llama [-0.72, $+0.45$], Qwen [-3.41, -1.64], Gemma [+1.79, +2.42], Mistral [-0.55, $+1.34$]. The two judges that agree best are Qwen2.5-32B and Llama-3.3-70B ($\kappa = 0.96$ on Llama). The pre-specified four-family test asked whether hedging follows popularity more than knowledge and entropy follows knowledge more than popularity, each on a fixed hedging readout: the regex on 24-token free replies for Llama (the reference, where both hold), Qwen and Mistral (only the entropy condition holds), and, because Gemma's replies rarely hedge, the forced-answer hedge log-odds for Gemma (both hold, although this readout shows no preference in Llama).

Permissive prompt (pre-specified). The user message was the question followed by "If you do not know the answer, say that you do not know.", on each model's existing two-judge question set, with three replies scored by the three judges and a majority rule. Cell rates (obscure unknown, obscure known, popular unknown, popular known): Llama 0.84, 0.69, 0.22, 0.00; Gemma 0.54, 0.41, 0.11, 0.00. Qwen2.5-7B, Mistral-7B and OLMo-2-7B-Instruct

Table 3: Judge agreement and abstention. Top: judge agreement and regex recall. Middle: abstention in the obscure, unknown cell. Bottom: three-judge popularity preference, then effects on first-token hedge-versus-answer log-odds at the forced-answer position, and the pre-specified four-family test, whose two conditions are that hedging follows popularity more than knowledge on this readout and that entropy follows knowledge more than popularity.

	Llama	Qwen	Gemma	Mistral
Two judges, κ (bar 0.60)	0.62	0.30	0.47	0.39
Three judges, Fleiss κ (bar 0.60)	0.74	0.41	0.64	0.54
Regex recall against two judges	0.99	0.58	0.48	0.73
Both judges DONT_KNOW	0.48	0.05	0.05	0.10
Both judges DONT_KNOW or CLARIFY	0.65	0.32	0.22	0.25
Three judges, majority	0.75	0.10	0.11	0.19
Regex detector	0.85	0.18	0.07	0.21
Three judges: popularity − knowledge	+0.44	–	+0.01	–
Hedge log-odds: popularity	+2.93	+1.14	+1.79	+2.48
Hedge log-odds: knowledge	+3.05	+3.63	−0.32	+2.08
Hedge log-odds: popularity − knowledge	−0.12	−2.49	+2.11	+0.40
Four-family test (pre-specified): conditions met	reference	one holds	both hold	one holds

(pre-specified; OLMo on its own pool, obscure known $n = 48$) are unmeasured under the same rule (Fleiss κ 0.40, 0.50, 0.46); their point estimates are +0.15 [+0.00, +0.30], +0.16 [+0.05, +0.27] and −0.09 [−0.22, +0.03]. The criterion let Gemma's verdict replace "cannot be estimated" only if its obscure unknown rate reached 0.25.

C Steering details

Pre-specified (Appendix G). Directions were fitted on a random half of each model's questions at layer round$(0.55 \times (L - 1))$, the last position of the free prompt: obscure minus popular and unknown minus known (differences of cell means, each averaged over the other factor), and abstain minus answer (questions whose stored free-generation regex abstention rate is at least 0.5 minus those with rate 0). Their natural norms are 4.8, 3.1 and 7.0 in Llama and 62.4, 66.5 and 105.3 in Gemma. The obscure-minus-popular direction with its abstention component projected out was rescaled to the original norm. Each direction was added at a dose $k \in \{0.5, 1, 2, 4, 8\}$ times the obscure-minus-popular direction's natural norm, at the last prompt position and every generated position. k was chosen on 150 popular known questions as the smallest value at which that direction beat five random directions by 0.10 in regex abstention, a rule separate from the confirmatory bar (1× for Llama, 4× for Gemma, whose obscure-minus-popular regex rate was 0.00 at 1× and 0.02 at 2×). All conditions were scored on 150 further popular known questions, 21 of which are also in the two-judge sample.

Table 4: Share of steered replies both judges label "I don't know" (regex detector in parentheses), 150 evaluation questions, 3 replies each.

	None	Random	Unk.−known	Abstain−answer	Obscure−popular	... abstention removed
Llama	0.01 (0.03)	0.008	0.04 (0.10)	0.09 (0.25)	0.10 (0.23)	0.02 (0.02)
Gemma	0.00 (0.00)	0.003	0.04 (0.03)	0.20 (0.74)	0.04 (0.27)	0.02 (0.07)

Primary scorer (both judges), obscure-minus-popular minus random: Llama +0.09 [+0.06, +0.12], Gemma +0.04 [+0.02, +0.06]; with abstention removed: +0.01 [−0.00, +0.02] and +0.02 [+0.01, +0.03]; obscure-minus-popular minus unknown-minus-known: +0.06 [+0.02, +0.09] and +0.00 [−0.03, +0.03]. Secondary scorer (a majority of three judges, added by amendment before judging): obscure-minus-popular minus random +0.17 [+0.13, +0.22] and +0.11 [+0.08, +0.15]; with abstention removed +0.02 [+0.00, +0.04] and +0.05 [+0.03, +0.07]. For Gemma the secondary scorer is the defensible one, since its two judges agree below the bar. The cosine between the obscure-minus-popular and abstention directions is 0.95 in Llama and 0.63 in Gemma. No condition produced degenerate text.

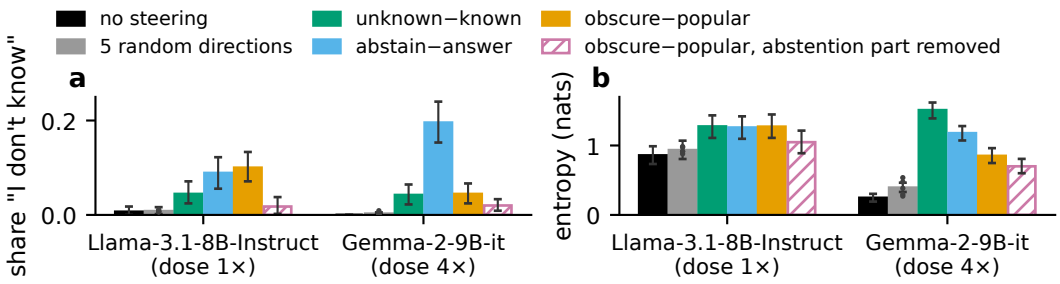


Figure 2: **In Llama, steering along the obscure-minus-popular direction adds little abstention beyond its overlap with an abstention direction.** (a) Share of free replies both judges label "I don't know" on 150 held-out popular questions each model knows; dots: random directions; hatched: obscure-minus-popular with its abstain-minus-answer component projected out. Gemma's two judges agree below the bar; three-judge differences are given in the text below. (b) First-token entropy under the forced-answer prompt. Error bars: 95% bootstrap over questions.

Within-level abstention direction (pre-specified). To remove the built-in overlap, the abstain-minus-answer direction was refitted on obscure questions only (587 abstaining against 55 answering in Llama, 36 against 520 in Gemma), with the same splits, layer and dose, and scored by a majority of three judges. Its cosine with obscure-minus-popular is 0.91 in Llama and 0.47 in Gemma. Rates for no steering, the random mean, obscure-minus-popular, the within-level abstention direction and obscure-minus-popular with that direction removed: Llama 0.02, 0.02, 0.19, 0.22, 0.03; Gemma 0.00, 0.01, 0.11, 0.41, 0.03. Over random directions, obscure-minus-popular adds +0.17 [+0.12, +0.22] and +0.10 [+0.07, +0.14], and with the within-level component removed +0.01 [+0.00, +0.03] and +0.03 [+0.01, +0.04]; by Llama's pre-specified rule the effect is carried by abstention.

Popular-arm abstention direction (pre-specified). To test whether the overlap is with abstention in general, the abstain-minus-answer direction was refitted on popular questions only (63 abstaining against 548 answering), with 20 random directions instead of 5, and scored by a majority of three judges (Llama, dose 1). Its cosine with obscure-minus-popular is 0.89. Over random directions it adds +0.14 [+0.10, +0.19], obscure-minus-popular adds +0.17 [+0.12, +0.21], and obscure-minus-popular with this direction removed adds +0.01 [−0.00, +0.02]: the obscure-minus-popular direction adds no abstention beyond its overlap with an abstention direction fitted without any obscure entities.

D Lens: window effects, the instruct models and onsets

Pre-specified. Hedge term: log-sum-exp of the lens log-probabilities of the hedge tokens " not", " I", " unknown", " unclear" and " no"; the full readout subtracts the same quantity for the answer-opening tokens " a", " the", " that", " in", " Yes" and " yes". Window: relative depth $\ell/(L-1) \in [1/3, 2/3]$. Intervals: 2,000 within-cell bootstraps. Each model was labeled known or unknown on its own prompt with the same sampling rule.

Table 5: Mid-depth window effects in base models, with 95% intervals.

	Full readout		Hedge term alone	
Model	Popularity	Knowledge	Popularity	Knowledge
Llama-3.1-8B	+0.67 [+0.59, +0.75]	+0.36 [+0.28, +0.44]	+0.55 [+0.46, +0.64]	+0.56 [+0.47, +0.65]
Gemma-2-9B	+0.39 [+0.34, +0.43]	+0.37 [+0.33, +0.42]	+0.28 [+0.21, +0.35]	+0.30 [+0.23, +0.37]
Gemma-2-2B	+0.40 [+0.33, +0.47]	+0.11 [+0.04, +0.18]	+0.10 [+0.03, +0.17]	+0.37 [+0.30, +0.43]
OLMo-3-7B	+0.29 [+0.19, +0.38]	−0.30 [−0.40, −0.20]	+0.61 [+0.53, +0.70]	+0.53 [+0.45, +0.62]

Instruct partners (pre-specified). The same readout, window and labeling rule were applied to Llama-3.1-8B-Instruct, Gemma-2-9B-it and Gemma-2-2B-it with their own prefitted lenses and chat-template forced-answer prompts. The criterion allowed the claim that instruction

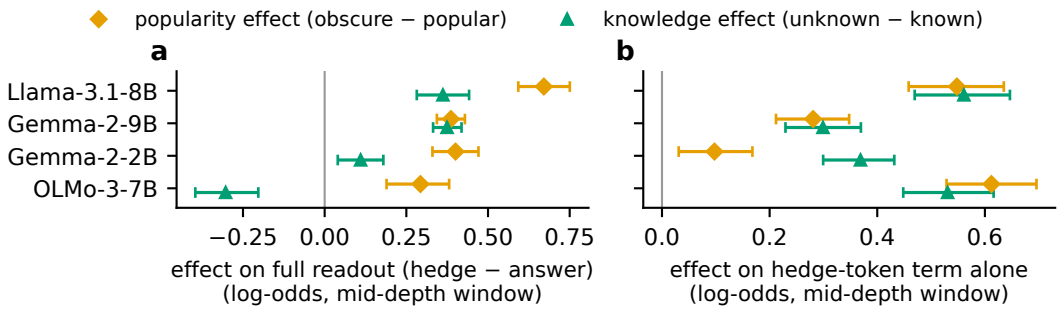


Figure 3: **In base models, the decoded hedge-token term responds to popularity and about as much to knowledge.** Jacobian-lens readout at the last position of "Question: ... Answer: The answer is", averaged over the pre-specified mid-depth window: (a) log-odds of five hedge tokens (e.g. " not", " unknown") against six answer-opening tokens (e.g. " the", " a"); (b) the hedge-token term alone. On (a), popularity exceeds knowledge in three models because the answer-token term also moves; on (b), knowledge is about as large or larger, except in OLMo, where it is slightly smaller with overlapping intervals. Points: popularity and knowledge effects with 95% bootstrap intervals. Lenses are fitted per model, so magnitudes are not comparable across models.

tuning increases the popularity preference only if, for a pair, the instruct difference exceeded the base difference with non-overlapping intervals; this held for no pair.

Table 6: Popularity minus knowledge in the mid-depth window, base and instruct, with 95% intervals. Lenses differ between the two models of a pair, so compare signs, not magnitudes.

	Full readout		Hedge term alone	
Pair	Base	Instruct	Base	Instruct
Llama-3.1-8B	+0.31 [+0.16, +0.46]	−0.35 [−0.71, +0.01]	−0.01 [−0.17, +0.15]	−1.46 [−1.99, −0.97]
Gemma-2-9B	+0.01 [−0.06, +0.08]	−0.98 [−1.17, −0.78]	−0.02 [−0.15, +0.10]	−1.28 [−1.56, −1.00]
Gemma-2-2B	+0.29 [+0.16, +0.42]	+0.15 [+0.07, +0.22]	−0.27 [−0.38, −0.15]	−0.89 [−1.02, −0.76]

Onsets. The onset of an effect curve is the first layer where it reaches half its positive maximum over layers and its 95% interval excludes zero. A model supports the prediction that familiarity acts at a shallower depth than knowledge, our layer-depth analogue of a timing account of feelings of knowing [Koriat and Levy-Sadot, 2001], if the knowledge onset follows the popularity onset with an interval above zero, and contradicts it if the interval lies below zero. The prediction was not supported (mixed): 1 supporting, 1 contradicting and 6 undecided of the eight base and instruct models.

Table 7: Lens onsets (layer index) of the popularity (F) and knowledge (K) effects, their difference $D = K - F$ with its 95% interval, the outcome, and the mid-depth popularity effect on the full readout. Instruct rows use the instruct models' own lenses. Onsets are integer layers, so bootstrap intervals for D can collapse to a point (two of eight here); they are the inputs to the pre-specified rule, not a precision estimate.

Model	Layers	F onset	K onset	D	Outcome	Mid popularity
Llama-3.1-8B-Instruct	32	17	17	0 [0, 0]	undecided	+1.58
Llama-3.1-8B	32	16	16	0 [0, +1]	undecided	+0.67
Qwen2.5-7B-Instruct	28	20	20	0 [−1, +1]	undecided	+0.21
Gemma-2-9B-it	42	24	24	0 [−2, +1]	undecided	+0.71
Gemma-2-9B	42	24	25	+1 [+1, +2]	supports	+0.39
Gemma-2-2B-it	26	17	15	−2 [−2, 0]	undecided	+0.40
Gemma-2-2B	26	16	16	0 [0, +1]	undecided	+0.40
OLMo-3-7B	32	19	14	−5 [−5, −5]	contradicts	+0.29

E Cross-model agreement (exploratory)

Spearman correlations across questions between Llama's and each other model's forced-answer hedge log-odds and first-token entropies, on questions where both models have the same knowledge label and where they differ (95% bootstrap intervals). The plan, including the rule that hedge agreement and the hedge-minus-entropy gap both have 95% intervals excluding zero, was written before this run, but the point estimates had been seen, so the analysis is post hoc.

	Llama–Qwen	Llama–Gemma	Llama–Mistral
Questions, knowledge differs	313	60	108
Hedge log-odds agreement	+0.11 [−0.00, +0.22]	+0.38 [+0.13, +0.60]	+0.25 [+0.06, +0.42]
Entropy agreement	−0.17 [−0.28, −0.07]	+0.03 [−0.24, +0.28]	−0.07 [−0.27, +0.12]
Gap (hedge − entropy)	+0.28 [+0.14, +0.42]	+0.35 [−0.03, +0.74]	+0.32 [+0.06, +0.57]
Rule	not met	not met	met
Questions, same knowledge	1,784	1,977	2,000
Hedge log-odds agreement	+0.50	+0.38	+0.64
Entropy agreement	+0.46	+0.54	+0.69

F A model-side familiarity measure (negative result)

Pre-specified. We computed the mean per-token surprisal of each subject name under the base Llama-3.1-8B in the context "Here is a name:". It is uncorrelated with log page views (Spearman −0.001). With knowledge and name length held fixed, its coefficient per standard deviation is +0.013 [−0.026, +0.059] on two-judge abstention and +0.07 [+0.05, +0.10] on the regex measure; adding the popularity arm moves the regex coefficient to +0.01 [−0.01, +0.03], although surprisal hardly correlates with the arm (point-biserial +0.02; −0.06 given relation and knowledge), which we have not resolved. Surprisal therefore does not stand in for popularity. Short names made of common words ("New York" as a book title) are a likely reason.

G Pre-specified analyses

Each pre-specified analysis followed a plan written before it ran, stating the measurement, the test and the wording each outcome would license. The plans are in the supplementary material, so each can be checked against the SHA-256 prefix we recorded before its run. The prefixes are self-recorded, not externally timestamped.

Analysis	Plan	SHA-256
Four-family design (Qwen)	`qwen_factorial_criterion.md`	eba35dec
Four-family design (Gemma, Mistral)	`gemma_mistral_factorial_criterion.md`	b53d9a27
One corrected judge	`hedge_judge_criterion.md`	ae0bb807
Two-judge measurement	`hedge_measure_v2_criterion.md`	81acfc9f
Lens study	`jlens_cascade_criterion.md`	c8e04372
Independent labels	`e1_independent_labels.md`	af50f53f
Name surprisal	`e2_name_surprisal.md`	2b69e06c
Lens split	`e3_lens_split.md`	b2b916ce
Steering, held-out dose	`e4_steering_v3.md`; amendment[a]	18c74883; 7bed3826
Third judge	`e5_third_judge.md`	f35aa4d9
Entity recognition	`e6_entity_knowledge.md`	937b46e3
Regraded labels, strict declines	`e7_label_and_decline_audit.md`	1498be91
Within-level steering	`e8_steering_within_level.md`	05474954
Permissive prompt	`e9_permissive_prompt.md`	f6459163
Instruct lens	`e10_instruct_lens.md`	6af179ce
Name-cue control	`e11_name_cue_control.md`	f716d396
Permissive prompt, Qwen and Mistral	`e12_permissive_qwen_mistral.md`	56831ebc
Fresh replies; new questions	`e13_e14_replications.md`	fbbc8c25
OLMo-2 prompt; popular-arm steering	`e15_e16_olmo_and_steering.md`	2b3e665d

[a] Adds the third judge as a secondary scorer; its internal date is one day later than its recorded hash time.

The guessability, namesake, name-spelling, free-reply, entropy-control and label-agreement checks, the alias audit and the cross-model agreement were decided after seeing data and are post hoc.